

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Report

Executive Summary

This report evaluates `qwen/qwen3.5-plus` across Infron and OpenRouter under prompt-caching workloads. The run uses only `/v1/chat/completions`, keeps model/platform default reasoning and thinking behavior, and keeps prompt-length stratification enabled.

After A/B filtering, the analysis retains **715 paired samples** and **2860 request-level observations**. It excludes **170 records**. Provider model IDs are Infron `qwen/qwen3.5-plus` and OpenRouter `qwen/qwen3.5-plus-20260420`.

Core finding: token-level cache hit rate ties in every routing mode. Infron leads observed cost and E2E latency in every routing mode. OpenRouter leads throughput and Streaming TTFT in every routing mode. Cache read/write telemetry is zero on both platforms in this run, so the report does not attribute a cache advantage to either side.

Quality Gate

Item	Value
Model	<code>qwen/qwen3.5-plus</code>
Provider model IDs	Infron <code>qwen/qwen3.5-plus</code> ; OpenRouter <code>qwen/qwen3.5-plus-20260420</code>
API protocol	<code>/v1/chat/completions</code> only
Groups / rounds	4 groups x 50 rounds
Prompt tiers	<code>short</code> ≈1500, <code>medium</code> ≈8000, <code>long</code> ≈32000 tokens
Reasoning / thinking	Default platform/model behavior; no explicit parameter
Pairing rule	strict sort/group/round pair with first/second <code>usage.prompt_tokens</code> deltas <= 50
Retained pairs	715
Request-level rows	2860
Excluded records	170

Route–Level Result Matrix

Routing mode	Pairs	Token cache hit rate	Observed cost	Throughput	E2E latency	TTFT
Throughput First	157	Tie 0.00% / 0.00%	Infron \$0.598793 / OpenRouter \$2.382274	Infron 3.491 / OpenRouter 50.324 tok/s	Infron 3481.37 ms / OpenRouter 27183.00 ms	Infron 3201.57 ms / OpenRouter 2634.81 ms
Price First	163	Tie 0.00% / 0.00%	Infron \$0.334323 / OpenRouter \$2.546479	Infron 2.875 / OpenRouter 49.998 tok/s	Infron 4243.01 ms / OpenRouter 27587.21 ms	Infron 3926.75 ms / OpenRouter 2840.22 ms
Latency First	200	Tie 0.00% / 0.00%	Infron \$1.228794 / OpenRouter \$3.092506	Infron 3.165 / OpenRouter 50.051 tok/s	Infron 3879.73 ms / OpenRouter 26905.27 ms	Infron 3577.67 ms / OpenRouter 2730.12 ms
TTFT First	195	Tie 0.00% / 0.00%	Infron \$1.210111 / OpenRouter \$3.009037	Infron 3.253 / OpenRouter 50.101 tok/s	Infron 3783.04 ms / OpenRouter 27125.24 ms	Infron 3453.44 ms / OpenRouter 2733.46 ms

API Protocol Record

API protocol	Endpoint	Platform	Planned pairs	Requests	Success rate	Usage coverage	Cost coverage	Cache telemetry coverage	HTTP status
chat_completions	/v1/chat/completions	Infron	800	1600	97.44%	100.00%	100.00%	100.00%	0:41, 200:1559
chat_completions	/v1/chat/completions	OpenRouter	800	1600	94.38%	100.00%	100.00%	100.00%	0:40, 200:1510, 429:50

Prompt–Length Stratified Results

Tier	Pairs	Token cache hit rate	Observed cost	E2E latency	TTFT	Reasoning tokens
short	238	Tie 0.00% / 0.00%	Infron \$0.133744 / OpenRouter \$1.309786	Infron 2604.15 ms / OpenRouter 23440.01 ms	Infron 2321.52 ms / OpenRouter 1968.25 ms	Infron 0 / OpenRouter 560458
medium	245	Tie 0.00% / 0.00%	Infron \$0.672494 / OpenRouter \$2.727598	Infron 3718.17 ms / OpenRouter 27129.94 ms	Infron 3397.07 ms / OpenRouter 2516.97 ms	Infron 0 / OpenRouter 662088

Tier	Pairs	Token cache hit rate	Observed cost	E2E latency	TTFT	Reasoning tokens
long	232	Tie 0.00% / 0.00%	Infron \$2.565783 / OpenRouter \$6.992912	Infron 5263.31 ms / OpenRouter 31074.85 ms	Infron 4943.35 ms / OpenRouter 3752.45 ms	Infron 0 / OpenRouter 697265

Reasoning / Thinking Observation

This run does not explicitly disable or force reasoning/thinking. Reasoning tokens are treated as observed telemetry.

Routing mode	Infron reasoning tokens	OpenRouter reasoning tokens	Infron avg/request	OpenRouter avg/request
Throughput First	0	423458	0.0000	1348.5924
Price First	0	443252	0.0000	1359.6687
Latency First	0	530769	0.0000	1326.9225
TTFT First	0	522332	0.0000	1339.3128

Reproducibility References

Artifact	Link	SHA256 / Notes
Chinese HTML report	< https://infronai.github.io/prompt-cache-bench/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/reports/routing-cache-cost-streaming-performance-ab-study__qwen3-5-plus__infron-vs-openrouter__4x50-stream-ttft-reasoning-default-length-tiers-chat-completions__2026-07-06.zh.html >	
English HTML report	< https://infronai.github.io/prompt-cache-bench/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/reports/routing-cache-cost-streaming-performance-ab-study__qwen3-5-plus__infron-vs-openrouter__4x50-stream-ttft-reasoning-default-length-tiers-chat-completions__2026-07-06.en.html >	

Artifact	Link	SHA256 / Notes
Summary	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/data/summary.json >	4f74247f769ca6ca3b71d26963e90b7daf1d571bcae1bba0144da487a
Paired dataset CSV	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/data/benchmark_pairs.csv >	aea743349fc3592d9a0f4a3fee4b03fea667eac00a09e2d1921d77973
Request-level dataset JSONL	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/data/benchmark_requests.jsonl >	cc6dcf3628aced3eb697c349cfbf18070c293b9333fc86204e344694b
Excluded-record audit	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/data/records_excluded.json >	995eb8dba3f1b33db3384304899ec076aa51f1c592a48ae22b839314e
Data directory	< https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/data >	
Benchmark runner source	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/code/rerun_routing_sort_cache_cost_ab.py >	b8ff71395fb08a6ff817c03d153ac09914b2bddb6994265a86a5ecaba

Artifact	Link	SHA256 / Notes
HTML report renderer source	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/code/render_glm52_deepseek_style_report.py >	d82c729d73b4a19087506bc2381fd1dcf1ccf0270160a064a6a4ed549
Test source	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/code/test_rerun_routing_sort_cache_cost_ab.py >	30b46e2b21db1b0e42899db12983d8e5bdfbaa8ceb9c64b04e5e664aa
A/B report standard	< https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/qwen/qwen3.5-plus/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-06/code/ab-report-standard.md >	4c8633866de695fafeec1f70477340877b89b9253f24eb6c06b367656

Dataset reference: `business_representative` built-in representative business templates; exported request-level dataset is `benchmark_requests.jsonl`.